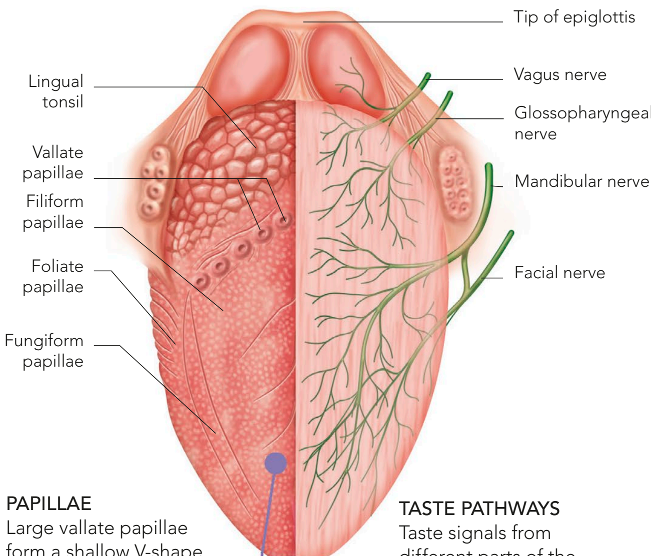
TASTE

Taste works in a similar way to smell. Its gustatory cell (taste) receptors detect specific chemicals dissolved in saliva by a "lock-and-key" method (see p.112). Groups of receptor cells are known as taste buds. A child has about 10,000 taste buds, but with age, their numbers may fall to under 5,000. They are located mainly on and between the pimple-like papillae that dot regions of the tongue's upper surface. There are also some taste buds on the palate (roof of the mouth), throat, and epiglottis.



TASTE BUDS

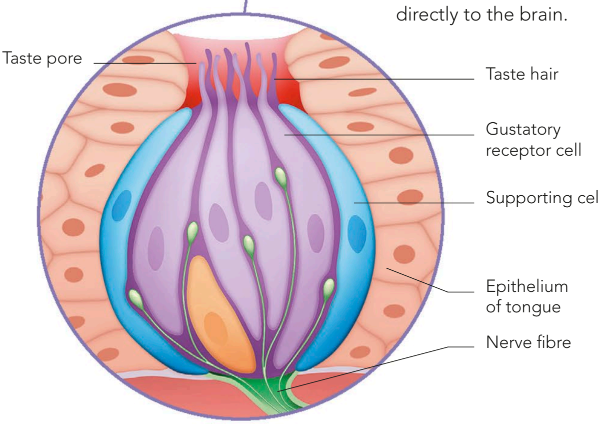
Each taste bud is structured much like an orange whose "segments" consist of roughly 25 "gustatory" receptor cells and numerous supporting cells. The receptor cells have hair-like tips that project into a hole (the taste pore) in the tongue's surface. Their nerve fibres gather at the bud base.

PAPILLAE

Large vallate papillae form a shallow V-shape at the rear of the tongue; the fungiform and filiform papillae are much smaller.

TASTE PATHWAYS

Taste signals from different parts of the tongue are conveyed by branches of three of the cranial nerves directly to the brain.



TASTE RECEPTORS

A scanning electron microscope image shows two different types of papillae. The purple conical structures are filiform papillae. The circular pink structure is a fungiform papilla.